

Q. If a regular expression is given as $(001)^*(01+10)$. Apply the rules to construct a regular grammar for this language. (7 Marks)

⇒ R.E. : $(001)^*(01+10)$

Rules to construct Regular Grammar from Regular Expression

① Rule for single symbol

if R.E is : a

then grammar is : $S \rightarrow a$

② Rule for concatenation

if R.E is : ab

then grammar is : $S \rightarrow aA$

$A \rightarrow b$

Ex R.E. : xyz

R.G. : $S \rightarrow xA$

$A \rightarrow yB$

$B \rightarrow z$

③ Rule for union (+)

if R.E. is : $a+b$

R.G. is : $S \rightarrow a|b$

Ex R.E. : $01+10$

R.G. : $S \rightarrow 0A|1B$

$A \rightarrow 1$

$B \rightarrow 0$

④ Rule for Kleene star (*)

R.E. : a^*

R.G. : $S \rightarrow aS|\Lambda$

⑤ Rule for positive closure (+)

R.E. : a^+

R.G. : $S \rightarrow aS|a$

⇒ Solve given Qn using rules

$(001)^*(01+10)$

① 001

$S \rightarrow 0A$

$A \rightarrow 0B$

$B \rightarrow 1$

② $(001)^*$

$S \rightarrow 0A$

$A \rightarrow 0B$

$B \rightarrow 1S$

③ $01+10$

$S \rightarrow 0C|1D$

$C \rightarrow 1$

$D \rightarrow 0$

④ Final Grammar

$S \rightarrow 0A|0C|1D$

$A \rightarrow 0B$

$B \rightarrow 1S$

$C \rightarrow 1$

$D \rightarrow 0$

Q. Find CFG (Context Free Grammar) for the language
 $L = \{a^i b^j c^k \mid j = i + k\}$ (4 Marks)

$\Rightarrow L = \{a^i b^j c^k \mid j = i + k\}$

$$j = i + k \rightarrow \text{no of } b\text{'s} = \text{no of } a\text{'s} + \text{no of } c\text{'s}$$

$\therefore$ we can write string as :

$$a^i b^{i+k} c^k = a^i b^i b^k c^k$$

$\rightarrow$ divide $a^i b^i$ $b^k c^k$ in two parts.

for $a^i b^i \rightarrow \Lambda, ab, aabb, aaabbb, \dots$

$$\therefore A \rightarrow aAb \mid \Lambda \quad \text{--- (1)}$$

for $b^k c^k \rightarrow \Lambda, bc, bbcc, bbbccc, bbbbcccc, \dots$

$$\therefore B \rightarrow bBc \mid \Lambda \quad \text{--- (2)}$$

Now, combine (1) & (2)

$$AB = a^i b^i b^k c^k$$

$$\therefore S \rightarrow AB \quad \text{--- (3)}$$

from (1), (2) & (3),

A CFG for $L = \{a^i b^j c^k \mid j = i + k\}$
is :

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid \Lambda$$

$$B \rightarrow bBc \mid \Lambda$$

Q. Find context-free grammars that generate the following languages : (4 Marks)

(i) $L_1 = \{a^i b^j c^k \mid i = j + k\}$

(ii) $L_2 = \{a^i b^j c^k \mid j = i \text{ or } j = k\}$

Q. write CFG for the following

(i) $L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$

(ii) $L = \{a^i b^j c^k \mid j > i + k\}$

(4 Marks)

Q. Find CFG generating the language of even-length strings in $\{a, b\}^*$ with two middle symbols equal. (4 marks)

Q. Construct the CFG for the language
 $L = \{x \in \{0, 1\}^* \mid n_0(x) \neq n_1(x)\}$ (7 marks)

Q. Find the CFG for the regular expression:
 $(01^*1 + 1)^*(01)^*$ (3 marks)

Q. Construct a CFG for set of strings that contain equal number of a's and b's over $\Sigma = \{a, b\}$
(3 marks)